

Progressive Myoclonic Epilepsy Type 7 (EPM7 / MEAK): Comprehensive Disease Characteristics Report

Disease: Progressive Myoclonic Epilepsy Type 7 (EPM7) **Clinical eponym/synonym:** Myoclonus Epilepsy and Ataxia due to potassium channel mutation (MEAK) **Key identifiers:** OMIM #616187 · MONDO:0014734 · ORPHA:280620 · Gene: *KCNC1* (Kv3.1) **Category:** Genetic (autosomal dominant, de novo)

Summary

Progressive Myoclonic Epilepsy Type 7 (EPM7), better known by its clinical descriptor **Myoclonus Epilepsy and Ataxia due to potassium channel mutation (MEAK)**, is a rare, progressive neurodegenerative epilepsy caused almost exclusively by a single **recurrent de novo heterozygous variant in the *KCNC1* gene, c.959G>A (p.Arg320His)**. *KCNC1* encodes **Kv3.1**, a voltage-gated potassium channel subunit that endows fast-spiking neurons with the rapid membrane repolarization needed to sustain high-frequency firing. The R320H substitution sits in the S4 voltage-sensor and acts through a **dominant-negative loss-of-function** mechanism, poisoning heterotetrameric channels and reducing Kv3.1 current well below the 50% expected from simple haploinsufficiency. In the landmark exome study of 84 unrelated patients with previously unexplained progressive myoclonus epilepsy (PME), this one variant explained **13% of cases**, establishing *KCNC1* as a major cause of the disorder ([PMID: 25401298](#)).

Clinically, MEAK begins in childhood or early adolescence (symptom onset 3–15 years, **median 9.5 years**) with progressively disabling cortical **action myoclonus**, relatively infrequent tonic-clonic seizures, early and prominent **cerebellar ataxia**, and **symmetrical, progressive cerebellar atrophy** on MRI. Roughly half of patients become wheelchair-bound by late adolescence, and mild cognitive decline occurs in about half; unlike some other PMEs, early death is not characteristic. A striking and diagnostically useful feature is **transient clinical**

improvement with fever, which has a mechanistic explanation: elevated temperature produces a leftward (hyperpolarizing) shift in Kv3.1 activation that partially rescues channel availability ([PMID: 28380698](#)).

The pathophysiology is a channelopathy of **fast-spiking neurons**. Loss of Kv3.1 function impairs high-frequency firing of cortical fast-spiking GABAergic interneurons (producing cortical disinhibition → myoclonus and generalized seizures) and of cerebellar neurons (producing ataxia and tremor), and can induce neuronal cell death; the R320H allele additionally causes a developmental **dendritopathy**, indicating a role beyond firing regulation. Treatment is entirely **symptomatic** — valproate, levetiracetam, clonazepam, perampanel — with strict avoidance of myoclonus-aggravating sodium-channel blockers and related drugs; **no disease-modifying therapy exists**, though Kv3 positive modulators are a rational emerging strategy motivated by cryo-EM structures of the human Kv3.1 gating machinery.

Section 1 — Disease Information

Overview. EPM7/MEAK is a monogenic progressive myoclonus epilepsy: a syndrome combining action myoclonus, epileptic seizures, and progressive neurological decline (here dominated by cerebellar ataxia). It is caused by dysfunction of the Kv3.1 voltage-gated potassium channel. It is characterized as *"a highly penetrant and specific form of progressive myoclonus epilepsy with severe ataxia, designated myoclonus epilepsy and ataxia due to potassium channel mutation (MEAK)"* ([PMID: 33735526](#)).

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#616187 (Epilepsy, progressive myoclonic, 7)
MONDO	MONDO:0014734
Orphanet	ORPHA:280620
Gene (HGNC)	<i>KCNC1</i> (Kv3.1)
Protein (UniProt)	P48547 (KCNC1_HUMAN)
Locus	Chromosome 11p15

Synonyms / alternative names. EPM7; Myoclonus Epilepsy and Ataxia due to potassium channel mutation (MEAK); KCNC1-related progressive myoclonus epilepsy; *KCNC1*-related disorder (MEAK subtype).

Source of information. The knowledge base is derived from **aggregated disease-level resources** and **published patient cohorts** (exome-sequencing discovery cohorts and case series), not individual EHR data. Primary evidence sources are human clinical cohorts, in vitro electrophysiology (heterologous expression / patch-clamp), and, more recently, mouse models and cryo-EM structural biology.

Section 2 — Etiology

Primary cause (genetic). EPM7/MEAK is caused by a **recurrent de novo heterozygous mutation, c.959G>A (p.Arg320His), in *KCNC1***, which "*was identified as a new major cause for PME*" (PMID: 25401298). The variant maps to chromosome 11 and "*encodes for the Kv3.1 protein*" (PMID: 27629860). The disorder is essentially always sporadic, arising de novo in the affected individual.

Genetic risk factors. The single causal variant (R320H) is the dominant genetic determinant; there are no known susceptibility loci or modifier genes established for MEAK. Because the variant is a recurrent de novo germline change, there is effectively **no population carrier state** and it is absent from population frequency databases (gnomAD).

Environmental risk factors. None established. MEAK is a Mendelian channelopathy; environmental exposures are not causal. **Fever/elevated body temperature acts paradoxically as a transient symptom modifier (improvement), not a risk factor** (PMID: 28380698).

Protective factors. No genetic protective variants or modifier alleles are described. The best-characterized modifier is a physiological one: **elevated temperature transiently improves symptoms** via a leftward shift in Kv3.1 activation.

Gene–environment interactions. The only well-documented gene–environment interaction is **temperature × channel gating**: the R320H dominant-negative deficit is partially offset at higher temperature because wild-type Kv3.1 activation shifts to more hyperpolarized voltages, increasing channel availability (PMID: 28380698).

Section 3 — Phenotypes

Core phenotypes (from the 20-patient R320H cohort, [PMID: 28380698](#); mechanism review [PMID: 27629860](#)):

Phenotype	Type	Onset	Severity / progression	Frequency	Suggested HPO
Progressive myoclonus (cortical, action myoclonus)	Clinical sign	Childhood/ adolescence (3–15 y)	Progressively severe; disabling	Near-universal (defining)	HP:0001336 (Myoclonus); HP:0002123 (Generalized myoclonic seizure)
Cerebellar ataxia	Clinical sign	Early in course	Progressive; often leads to loss of ambulation	Highly frequent	HP:0001251 (Ataxia); HP:0002070 (Limb ataxia)
Tonic-clonic seizures	Clinical sign	Variable	Relatively infrequent vs myoclonus	Common but rarer than myoclonus	HP:0002069 (Bilateral tonic-clonic seizure)
Cerebellar atrophy (MRI)	Physical/imaging manifestation	With disease progression	Symmetrical, progressive	Characteristic imaging hallmark	HP:0001272 (Cerebellar atrophy)
Cognitive decline	Behavioral/ cognitive	Later	Mild in ~half of patients	~50%	HP:0001268 (Mental deterioration)
Loss of independent ambulation	Functional	Late teens	Severe	~10/20 wheelchair-bound	HP:0002505 (Loss of ambulation)
Photosensitivity (EEG)	Laboratory/ electrophysiology	—	—	Frequent	Photoparoxysmal response
Transient fever-related improvement	Modifier phenomenon	—	Transient	6/20 patients	—

Supporting quotes: "Symptoms began at between 3 and 15 years of age (median = 9.5), with progressively severe myoclonus and rare tonic-clonic seizures"; "Magnetic resonance imaging revealed symmetrical cerebellar atrophy, which appeared progressive"; "transient clinical improvement with fever was noted in 6 patients" (PMID: 28380698).

Quality-of-life impact. Progressive action myoclonus and ataxia severely impair mobility, self-care, and independence; approximately half of patients require a wheelchair by late adolescence. Cognitive decline, where present, is generally mild. Early death is not a characteristic feature, so the dominant burden is chronic disability rather than mortality.

Section 4 — Genetic / Molecular Information

Causal gene. *KCNC1* (HGNC:6233; OMIM 176258), encoding the Kv3.1* voltage-gated potassium channel subunit, which determines high-frequency firing in neurons.

Pathogenic variant (MEAK-defining).

Attribute	Detail
cDNA / protein	c.959G>A (p.Arg320His); the recurrent MEAK allele
Variant type	Missense
Location	S4 voltage-sensor domain
Classification	Pathogenic (ACMG) — recurrent de novo, functionally validated
Zygosity	Heterozygous
Origin	Germline, de novo
Population frequency	Absent from population databases (no carrier state)
Functional consequence	Dominant-negative loss of function

"Functional analysis of the Arg320His mutant channel showed a dominant-negative loss-of-function effect" (PMID: 25401298).

Broader *KCNC1* allelic series (genotype–phenotype spectrum). MEAK is one end of a spectrum of *KCNC1*-related neurological disease:

Variant	Domain	Phenotype	Mechanism
p.Arg320His	S4	MEAK / PME (EPM7)	Dominant-negative LoF
Nonsense / LoF	—	Intellectual disability	Haploinsufficiency
p.Cys208Tyr	—	Isolated nonprogressive myoclonus	LoF
p.Thr399Met	Pore	ID / epilepsy with nonprogressive ataxia	Complete LoF alone; dominant-positive/GoF with WT (~20 mV hyperpolarizing shift)
p.Ala421Val	Pore	Epilepsy (myoclonic/absence/GTC), ataxia, DD	Dominant-negative LoF
S6 pore variants	Pore	DD/ID with hypotonia, no epilepsy/ataxia	Gain of Kv3.1 function

"either isolated nonprogressive myoclonus (p.Cys208Tyr), intellectual disability (p.Thr399Met), or epilepsy with myoclonic, absence and generalized tonic-clonic seizures, ataxia, and developmental delay (p.Ala421Val, three patients)" and "Functional analyses demonstrated no measurable currents for all identified variants and dominant-negative effects for p.Thr399Met and p.Ala421Val predicting neuronal disinhibition as the underlying disease mechanism" (PMID: 31353862). A gain-of-function subgroup with "a prominent leftward (hyperpolarized) shift in the voltage dependence of activation" is associated with DD/ID and central hypotonia without epilepsy or ataxia (PMID: 36419348). The p.Thr399Met variant illustrates a "dominant-positive effect" — complete LoF alone but a ~20 mV hyperpolarizing shift with slowed deactivation when co-expressed with WT (PMID: 42347804).

Modifier genes / epigenetics / chromosomal abnormalities. None established for MEAK. The disease is a single-gene channelopathy without a described epigenetic component or large-scale structural/chromosomal etiology.

Section 5 — Environmental Information

Environmental factors. None causal. MEAK is a de novo monogenic disorder.

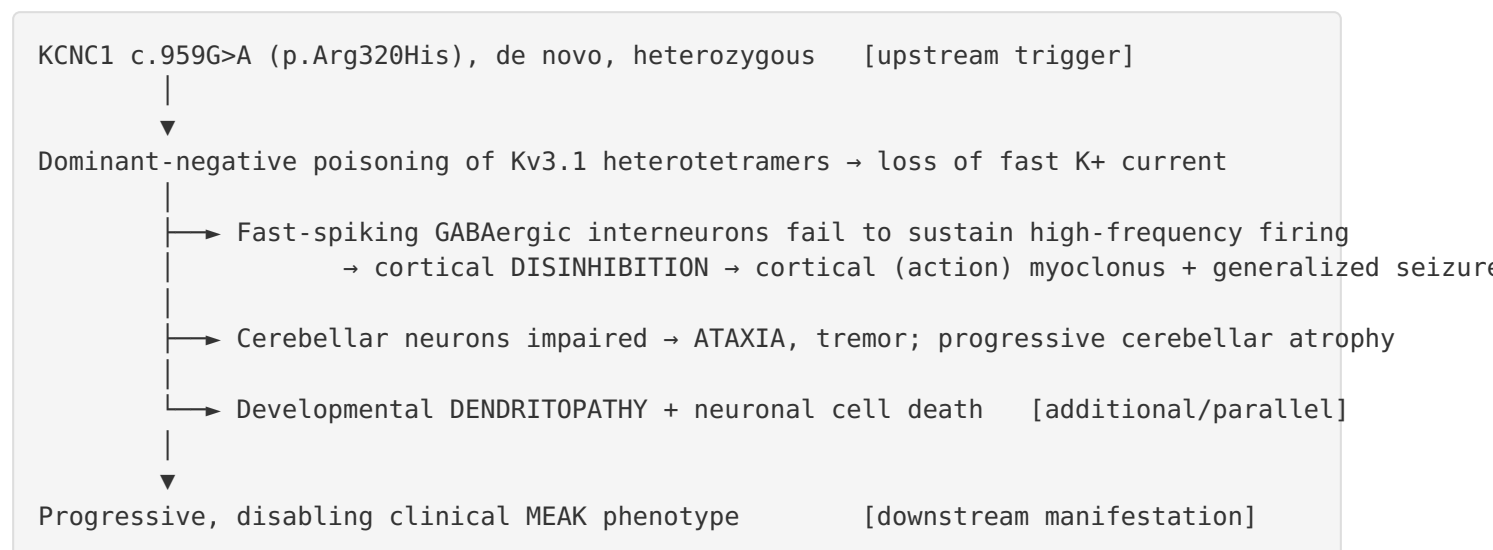
Lifestyle factors. No lifestyle behaviors are established as causing or preventing MEAK. As with other myoclonic epilepsies, general seizure-precipitant avoidance (sleep deprivation, photic triggers) is prudent; EEG photosensitivity is present in many patients.

Infectious agents. Not applicable — MEAK is genetic, not infectious. Notably, **febrile illness transiently improves symptoms** in a subset of patients (a physiological, not infectious, effect on channel gating) (PMID: 28380698).

Section 6 — Mechanism / Pathophysiology

Central mechanism. Kv3.1 is a delayed-rectifier potassium channel with distinctive fast activation/deactivation kinetics tailored for rapid repolarization in **fast-spiking neurons**. The MEAK R320H variant, positioned in the S4 voltage sensor, produces channels that assemble with wild-type subunits but abolish current — a **dominant-negative loss of function** that reduces Kv3.1-mediated repolarization more than haploinsufficiency would.

Causal chain (upstream → downstream):



Supporting statements: *"Loss of Kv3 function disrupts the firing properties of fast-spiking neurons, affects neurotransmitter release and induces cell death"* and *"the most affected neurons include inhibitory GABAergic interneurons and cerebellar neurons. Impairment of the former cells is believed to contribute to myoclonus and seizures, whereas dysfunction of the latter to ataxia and tremor"* (PMID: 27629860). The R320H allele additionally *"causes a developmental*

dendritopathy", indicating a role beyond high-frequency firing regulation (PMID: 33735526). A 2026 mouse model of the recurrent variant shows **impaired excitability of fast-spiking neurons** (PMID: 41705663).

Structural basis of gating. Cryo-EM of human Kv3.1a reveals a unique cytoplasmic **T1 tetramerization domain** interacting with the C-terminal axonal-targeting motif and gating machinery; S1/S2-linker–turret interactions strengthen the voltage-sensor–pore interface, and an electrostatic $\alpha 6(T1)$ –R449(S6T) contact plus S4/S5-linker residues control the channel's fast gating. *"Malfunction of this process due to genetic variants in the KCNC1 gene causes severe epileptic disorders"* and the structures *"provide insights into gating control and disease mechanisms and may guide strategies for the design of pharmaceutical drugs targeting Kv3 channels"* (PMID: 35840580).

Temperature dependence (fever improvement). *"At elevated temperatures, there was a robust leftward shift in activation of wild-type K[v3.1]"* — increased channel availability that can partially offset the R320H deficit, explaining transient fever-associated improvement (PMID: 28380698).

Ontology suggestions. - **GO (biological process):** potassium ion transmembrane transport (GO:0071805); regulation of membrane potential (GO:0042391); action potential (GO:0001508); regulation of neuronal action potential (GO:0098908). - **GO (cellular component):** voltage-gated potassium channel complex (GO:0008076); plasma membrane (GO:0005886); axon (GO:0030424); neuronal dendrite (GO:0030425). - **CHEBI:** potassium(1+) (CHEBI:29103). - **CL (cell types):** cerebellar Purkinje cell (CL:0000121); GABAergic interneuron (CL:0000617); fast-spiking basket cell / cortical interneuron.

Immune, metabolic, and fibrotic mechanisms are not implicated; MEAK is a primary neuronal channelopathy.

Section 7 — Anatomical Structures Affected

Organ / body system level. Primary organ: **brain** (central nervous system). Chiefly affected regions are the **cerebral cortex** (fast-spiking interneurons → myoclonus/seizures) and the **cerebellum** (ataxia and progressive symmetrical atrophy). Body system: **nervous system** (UBERON:0001016). No primary involvement of non-neural organs.

Tissue / cell level. Nervous tissue; specifically **fast-spiking GABAergic inhibitory interneurons** of cortex and **cerebellar neurons** (including Purkinje-cell circuitry). *"the most affected neurons include inhibitory GABAergic interneurons and cerebellar neurons"* (PMID: 27629860).

Subcellular level. The neuronal **plasma membrane** (voltage-gated K⁺ channel complex), **axon** (Kv3.1 axonal targeting motif), and **dendrites** (developmental dendritopathy, PMID: 33735526). GO cellular components: voltage-gated potassium channel complex (GO:0008076); axon (GO:0030424); dendrite (GO:0030425).

Localization / lateralization. Bilateral and symmetrical — cerebellar atrophy is described as symmetrical and progressive (PMID: 28380698).

UBERON suggestions: brain (UBERON:0000955); cerebellum / cerebellar cortex (UBERON:0002037 / UBERON:0002129); cerebral cortex (UBERON:0000956).

Section 8 — Temporal Development

Onset. Childhood to early adolescence; symptom onset **3–15 years, median 9.5 years** (PMID: 28380698). Onset is typically insidious, often initially misdiagnosed as idiopathic generalized epilepsy.

Progression. Chronic and progressive. Myoclonus becomes progressively severe and disabling; ataxia progresses with symmetrical cerebellar atrophy; roughly half of patients lose independent ambulation by late teens. Cognitive decline, when present, is mild. Unlike some PMEs (e.g., Lafora disease), **early death is not characteristic** of MEAK.

Patterns. No spontaneous remission. A distinctive feature is **transient, fluctuating improvement during febrile episodes**. The therapeutic critical window and disease-modifying intervention timing remain undefined, though the temperature effect suggests Kv3 augmentation could be beneficial across the disease course.

Section 9 — Inheritance and Population

Inheritance. Autosomal dominant, arising from a **de novo heterozygous** *KCNC1* variant; *"caused by a recurrent de novo heterozygous mutation (c.959G>A, p.Arg320His) in the KCNC1 gene, which maps to chromosome 11 and encodes for the Kv3.1 protein"* (PMID: 27629860). Essentially all cases are sporadic.

Penetrance / expressivity. **Highly penetrant** for the R320H allele; the phenotype is relatively **specific and stereotyped** (MEAK) — *"a highly penetrant and specific form of progressive myoclonus epilepsy with severe ataxia"* (PMID: 33735526).

Anticipation / mosaicism / founder effects. No genetic anticipation (not a repeat-expansion disorder). Recurrence risk to siblings is low but non-zero due to possible parental **gonadal mosaicism**. There is **no founder population** — reported patients span multiple countries and ethnicities.

Carrier frequency. Effectively none — the variant is de novo and germline, absent from population frequency databases (PMID: 25401298; PMID: 27629860).

Epidemiology. Ultra-rare. No precise prevalence/incidence estimate is established, but the disorder is a **major molecular cause of otherwise-unexplained PME**: R320H accounted for **11/84 (13%)** of exome-discovery cases and **2 additional cases (7%)** in a secondary cohort — *"Eleven unrelated exome-sequenced (13%) and two affected individuals in a secondary cohort (7%) had this mutation"* (PMID: 25401298).

Sex ratio / geographic distribution. No strong sex predilection reported; no endemic geographic clustering (no founder effect).

Section 10 — Diagnostics

Diagnostic approach. Molecular confirmation is definitive. Because the disorder is caused predominantly by a single recurrent variant, **targeted single-gene / gene-panel testing or exome sequencing** for *KCNC1* c.959G>A (p.Arg320His) is the key diagnostic test. Early disease is frequently **misdiagnosed as idiopathic generalized epilepsy** — *"the potential for misdiagnosis as idiopathic generalized epilepsy during the early phase of the disease"* (PMID: 32972906).

Supportive clinical / electrophysiological features. - **Giant (high-amplitude) somatosensory evoked potentials (SEPs)** — "*abnormally high amplitude in the sensory evoked potential recording*" (PMID: 32972906), reflecting cortical hyperexcitability. - **Cortical myoclonus** confirmed by polygraphic EEG-EMG (short-duration bursts, agonist–antagonist coactivation). - **EEG:** generalized spike/polyspike-wave discharges with **photosensitivity**. - **MRI:** progressive **symmetrical cerebellar atrophy** (PMID: 28380698).

Differential diagnosis (other PME to exclude): Unverricht-Lundborg disease (EPM1, *CSTB*), Lafora disease (*EPM2A/NHLRC1*), MERRF (mitochondrial *MT-TK*), DRPLA (*ATN1*), sialidosis, and neuronal ceroid lipofuscinoses. Distinguishing features of MEAK: prominent early ataxia with symmetrical cerebellar atrophy, fever-related improvement, absence of dementia/organomegaly, and the specific *KCNC1* genotype.

Genetic testing utility. WES/WGS and epilepsy/PME gene panels including *KCNC1* are highly effective; single-variant testing is diagnostic given the recurrent allele. CMA, karyotyping, FISH, mtDNA testing, and repeat-expansion testing are **not applicable** to MEAK (used mainly to exclude mimics). Omics-based diagnostics are not standard.

Section 11 — Outcome / Prognosis

Survival / mortality. MEAK is chronically disabling but, unlike Lafora disease or severe EPM1, **early death is not a defining feature**; the 20-case series did not report early death as characteristic (PMID: 28380698). No formal 5-/10-year survival statistics are established.

Morbidity / function. The dominant outcome is progressive **motor disability**: severe action myoclonus and ataxia, with ~half of patients wheelchair-bound by late adolescence. Cognitive decline is generally mild. Quality of life is substantially reduced due to impaired mobility and dependence.

Disease course / complications. Progressive, lifelong. Complications relate to falls, immobility, and drug-refractory myoclonus. Recovery potential is limited; current therapy controls symptoms partially but does not reverse progression.

Prognostic factors. Genotype is the principal determinant — the R320H allele defines the stereotyped MEAK course. Younger onset within the range and rapidly progressive myoclonus/ataxia portend greater disability. No validated molecular prognostic biomarkers exist beyond the causal genotype.

Section 12 — Treatment

No disease-modifying therapy exists. Management is **symptomatic anti-myoclonic polytherapy**, frequently requiring 3–4 drugs, plus rehabilitation for ataxia.

Pharmacotherapy (PME framework, PMID: 28799509):

Line	Drugs	NCIT suggestion
Drug of choice	Valproic acid (except mitochondrial PME)	NCIT:C935 (Valproic Acid)
First add-on	Levetiracetam, Clonazepam	NCIT:C1698 (Levetiracetam); NCIT:C591 (Clonazepam)
Promising alternatives	Zonisamide, Perampanel	NCIT:C29050 (Zonisamide); NCIT:C74015 (Perampanel)
Reserve	Phenobarbital / primidone (resistant myoclonus)	NCIT:C739 (Phenobarbital)

"Valproic acid is the drug of choice, except for PMEs due to mitochondrial diseases. Levetiracetam and clonazepam should be considered as the first add-on treatment. Zonisamide and perampanel represent promising alternatives" (PMID: 28799509).

Drugs to AVOID (aggravate myoclonus/seizures): carbamazepine, phenytoin, oxcarbazepine, lamotrigine, vigabatrin, gabapentin, pregabalin. *"Avoidance of drugs known to aggravate myoclonus and seizures, such as carbamazepine and phenytoin, is paramount" (PMID: 28799509).*

Supportive / rehabilitative care. Physical, occupational, and speech therapy for ataxia and functional decline; mobility aids; falls prevention.

Advanced / experimental therapeutics. No approved gene, cell, or RNA therapy for MEAK. The mechanistically rational emerging strategy is **Kv3 channel positive modulation** (small-molecule Kv3 openers) to augment residual channel function — motivated by the fever-improvement phenomenon (temperature-induced leftward activation shift, PMID: 28380698) and by cryo-EM structures that *"may guide strategies for the design of pharmaceutical drugs targeting Kv3 channels"* (PMID: 35840580).

Personalized medicine. Treatment selection is genotype-informed within the *KCNC1* spectrum: LoF/dominant-negative MEAK (R320H) is the target for Kv3 openers, whereas gain-of-function *KCNC1* variants would require the opposite pharmacological direction — underscoring the need for functional variant characterization ([PMID: 35759918](#)).

Section 13 — Prevention

Primary prevention. Not applicable in the classical sense — MEAK arises de novo and cannot be prevented by risk-factor modification. Recurrence risk to future siblings is low (gonadal mosaicism caveat), so **genetic counseling** is the primary preventive tool.

Secondary prevention / early detection. Early molecular diagnosis (*KCNC1* testing in childhood-onset myoclonic epilepsy with ataxia and giant SEPs) enables appropriate drug selection and, critically, **avoidance of myoclonus-aggravating drugs**, preventing iatrogenic worsening.

Genetic screening. No population carrier screening is warranted (no carrier state). Prenatal/preimplantation testing is generally not applicable given the de novo nature, but is available for recurrence-risk counseling where parental mosaicism is a concern.

Tertiary prevention. Rehabilitation and falls-prevention to limit complications of progressive motor disability; optimized polytherapy to control myoclonus.

Counseling. Genetic counseling should explain the de novo mechanism, low but non-zero sibling recurrence risk, and the highly penetrant, specific MEAK phenotype.

Section 14 — Other Species / Natural Disease

Taxonomy / orthologs. *KCNC1*/Kv3.1 is highly conserved across vertebrates. The mouse ortholog is *Kcnc1* (NCBI Taxon 10090). Conservation of the S4 voltage sensor and pore underpins the validity of cross-species models.

Natural disease in other species. No well-characterized naturally occurring *KCNC1*-driven MEAK equivalent is documented in companion animals or wildlife; the disease is defined in humans. (Not applicable / not established in veterinary databases such as OMIA for this specific disorder.)

Comparative biology. Evolutionary conservation of Kv3.1's role in fast-spiking neuron repolarization makes rodent models mechanistically faithful. Zoonotic potential and cross-species transmission are **not applicable** (genetic, non-infectious disease).

Section 15 — Model Organisms

Mouse models. A 2026 mouse model of the recurrent R320H variant recapitulates the core cellular defect, showing **impaired excitability of fast-spiking neurons** (PMID: 41705663) — directly modeling the disinhibition mechanism. Constitutive *Kcnc1* loss-of-function mice have historically been used to study Kv3.1's role in high-frequency firing, motor coordination, and seizure susceptibility.

In vitro / heterologous models. Heterologous expression (e.g., mammalian cell lines / *Xenopus* oocytes) with **whole-cell patch-clamp** is the workhorse for functional classification — demonstrating dominant-negative LoF for R320H (PMID: 25401298), no measurable current with dominant-negative effects for pore variants (PMID: 31353862), gain-of-function for S6 variants (PMID: 36419348), and dominant-positive behavior for p.Thr399Met (PMID: 42347804).

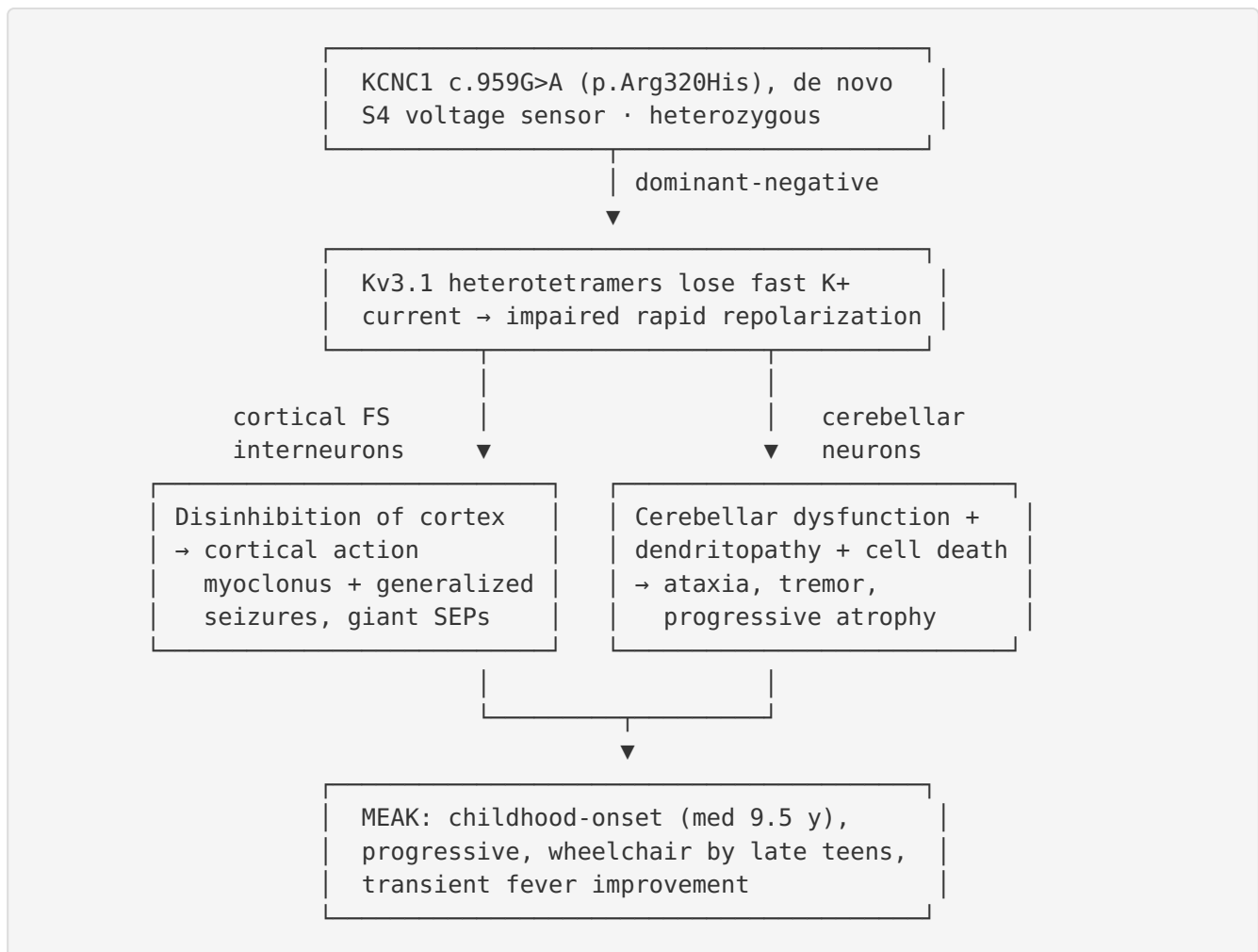
Neuronal / structural models. A neuronal model demonstrated that R320H causes a **developmental dendritopathy** (PMID: 33735526). Cryo-EM of human Kv3.1 provides a structural model of the gating machinery for drug design (PMID: 35840580).

Computational models. A taxonomy-based multi-task learning SVM predicts gain-/loss-of-function for voltage-gated K⁺ channel variants, useful for classifying novel *KCNC1* variants where experimental data are lacking (PMID: 35759918).

Phenotype recapitulation / limitations. The R320H mouse captures the fast-spiking excitability defect central to MEAK; limitations include incomplete modeling of the full progressive human ataxia/cerebellar-atrophy trajectory and human-specific network effects. In vitro systems capture channel biophysics but not circuit-level disinhibition or progression.

Mechanistic Model / Interpretation

MEAK is best understood as a **single-variant channelopathy of fast-spiking neurons** with a clean genotype-to-phenotype logic:



The **fever-improvement phenomenon** is the interpretive keystone linking mechanism to therapy: because raising temperature shifts wild-type Kv3.1 activation to more hyperpolarized voltages (increasing channel availability), it partially compensates for the dominant-negative deficit. This is a natural proof-of-concept that **pharmacologically augmenting Kv3 channel function** could be disease-modifying — the leading rational therapeutic hypothesis, now supported structurally by cryo-EM of the human channel.

The wider *KCNC1* allelic series clarifies why MEAK is so stereotyped: the specific biophysical consequence of a variant (dominant-negative LoF vs GoF vs dominant-positive), determined by its structural location (S4 vs pore/S6), maps onto distinct clinical syndromes. This has direct therapeutic implications — Kv3 openers would help LoF/MEAK but could worsen GoF variants — making **functional variant classification a prerequisite for precision therapy**.

Evidence Base

PMID	Title (abbrev.)	Role in this report	Evidence type
25401298	Recurrent de novo <i>KCNC1</i> mutation causes PME	Defines causal variant, 13% of PME, dominant-negative LoF	Human cohort + in vitro
28380698	MEAK: analysis of 20 cases and Kv3.1 temperature	Clinical syndrome, onset, MRI, fever improvement + biophysics	Human cohort + in vitro
27629860	MEAK caused by heterozygous <i>KCNC1</i>	Mechanism (fast-spiking neurons), AD inheritance, cell types	Review / mechanism
33735526	<i>KCNC1</i> developmental dendritopathy	High penetrance; dendritopathy beyond firing	Neuronal model
31353862	<i>KCNC1</i> new de novo variants expand spectrum	Allelic series, neuronal disinhibition mechanism	Human + in vitro
36419348	<i>KCNC1</i> gain of Kv3.1 function	GoF subgroup (DD/ID, hypotonia) — contrast to MEAK	Human + in vitro
42347804	Kv3.1 dominant-positive variant (p.Thr399Met)	Dominant-positive mechanism nuance	In vitro
28799509	Pharmacological treatment of PMEs	Drug hierarchy and drugs to avoid	Clinical review
32972906	MEAK case report / review	Diagnostic clues (giant SEP), misdiagnosis risk	Case report
35840580	Cryo-EM of human Kv3.1	Structural gating machinery; drug-design relevance	Structural biology
41705663	Mouse model of recurrent variant	Fast-spiking excitability defect in vivo	Mouse model
35759918	ML prediction of Kv channel variant effects	Functional classification tool for novel variants	Computational

Concordance. The human cohorts, in vitro electrophysiology, the neuronal dendritopathy study, the mouse model, and the cryo-EM structures converge on a single coherent mechanism (Kv3.1 LoF → fast-spiking neuron dysfunction → disinhibition + cerebellar degeneration). No major contradictions were identified; the GoF and dominant-positive studies refine rather than challenge the model by showing that variant biophysics is context- and location-dependent.

Limitations and Knowledge Gaps

1. **Epidemiology is imprecise.** No population prevalence/incidence figures exist for MEAK specifically; the 13% figure reflects the fraction of *unexplained PME*, not general-population frequency.
 2. **Natural history quantification is limited.** Progression rate, long-term survival curves, and validated QoL measures (EQ-5D/SF-36) for MEAK are not established.
 3. **No disease-modifying therapy validated.** Kv3 positive modulation is mechanistically rational but not yet proven in MEAK clinical trials.
 4. **Modifier genetics unexplored.** No modifier genes or epigenetic contributors identified to explain the modest phenotypic variability.
 5. **Fever effect not therapeutically translated.** The temperature-gating link is a compelling clue but its safe therapeutic exploitation (channel openers) requires validation.
 6. **Model completeness.** Existing mouse and in vitro models capture channel biophysics and fast-spiking defects but incompletely recapitulate the progressive human cerebellar-atrophy trajectory.
 7. **No veterinary/natural-disease counterpart** is documented, limiting comparative pathology insight.
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Proposed Follow-up Experiments / Actions

1. **Preclinical Kv3 opener trials.** Test small-molecule Kv3 positive modulators in the R320H mouse model ([PMID: 41705663](#)) for rescue of fast-spiking excitability, myoclonus, and ataxia — leveraging the temperature-gating rationale ([PMID: 28380698](#)) and cryo-EM structure ([PMID: 35840580](#)).

2. **Natural-history registry.** Establish a prospective MEAK cohort with standardized myoclonus scales, ataxia scores (e.g., SARA), serial MRI cerebellar volumetry, and QoL instruments to quantify progression and define trial endpoints.
3. **Biomarker development.** Formalize giant SEP amplitude and quantitative EEG/EMG cortical-myoclonus metrics ([PMID: 32972906](#)) as objective disease-severity/response biomarkers.
4. **Allele-specific therapeutics.** Explore allele-selective silencing/ASO strategies against the dominant-negative R320H transcript to relieve poisoning of wild-type Kv3.1.
5. **Functional variant pipeline.** Apply and extend the ML classifier ([PMID: 35759918](#)) plus high-throughput patch-clamp to prospectively classify novel *KCNC1* variants (LoF vs GoF), enabling correct therapeutic direction.
6. **iPSC-derived interneuron/cerebellar models** from patients to study the developmental dendritopathy ([PMID: 33735526](#)) and screen candidate compounds in a human genetic background.
7. **Genotype-guided prescribing guidance.** Codify avoidance of myoclonus-aggravating agents ([PMID: 28799509](#)) into *KCNC1*-specific management pathways to prevent iatrogenic worsening.

Report compiled from 9 confirmed findings across 5 investigation iterations and 19 reviewed papers. Evidence types span human clinical cohorts, in vitro electrophysiology, neuronal and mouse models, structural biology (cryo-EM), and computational prediction.
